A Cost-and-Behavior Case Study of Two GLEE Agents with Design-Time and Runtime Behavioral Interpretation

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Abstract

We present a descriptive cost-and-behavior case study of two *unmatched* deployments in the GLEE competition, which operationalize behavior-derived lessons at opposite ends of the stack. Agent A compiles conclusions from its match logs into rules at *design time*: game-theoretic solvers, revised offline by an AI assistant, no LLM calls at runtime. Agent B consumes lessons distilled from its own logs at *runtime*: an LLM makes every decision, briefed by a one-page lessons block. The observed B−A bargaining difference was +0.8 percentage points; the negotiation and persuasion point estimates were also close. The cost structures differed sharply: A's interpretation was run-level design-time expenditure ($151) followed by sub-millisecond decisions with no model-API expenditure, while B's decisions ran through a per-decision LLM layer ($1,158; $0.28 per game at 5.7 s per move). Similar payoff point estimates came from visibly different behavior: B settled bargains earlier, traded negotiation margin for closing rate, and lied at roughly 40% of A's designed lying rate. We offer these joint measurements as hypothesis-generating groundwork for controlled comparisons of where behavioral interpretation enters an agent.

1 Introduction

Beyond benchmark scores, the Interpreting Agent Behavior program asks what agents actually *did* [3]. We study two entries in GLEE [11], a competition where agents and humans play bargaining, negotiation, and persuasion games, and LLMs show both strategic competence and human-like biases [1, 5, 7]. Our agents consume behavior-derived lessons at different stages:

Agent A (design time) runs solver-based rules with pragmatic clamps and data-informed revisions: an AI assistant read the accumulating match logs offline, interpreted rejection patterns, and edited the rules. Interpretation was *compiled*, runtime deterministic. **Agent B (runtime)** sends every decision to an LLM; conclusions distilled from B's own logs by a deterministic template pipeline ride along in every call as a `<lessons>` block. Interpretation was *prompted*.

This is an observational comparison, not a controlled experiment: beyond where interpretation enters, the two systems differ in decision mechanisms, distillation mechanisms, iteration counts, game volumes ($\sim 29\times$), potentially matchmaking pools and periods, and base strategies (§2). Related work has studied the pieces separately: policy distillation compresses learned policies into cheaper executables [9], test-time-compute studies trade inference budget against quality on static benchmarks [10], and LLM-agent studies characterize strategic behavior in economic games [1, 5, 11]. We jointly account for behavior, policy assurance, latency, and model-API-equivalent expenditure to generate hypotheses rather than identify effects. We ask: **what did each deployment return, and what did it cost, in these two competition runs?**

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2 Two agents, one interpretation loop

Figure 1 contrasts the two pipelines (competition entries `Rubinstein` and `Rubinstein-RT`): both run a similar loop (trajectories in, conclusions out), but the interpretation enters above the design-time/runtime line in A and below it in B.

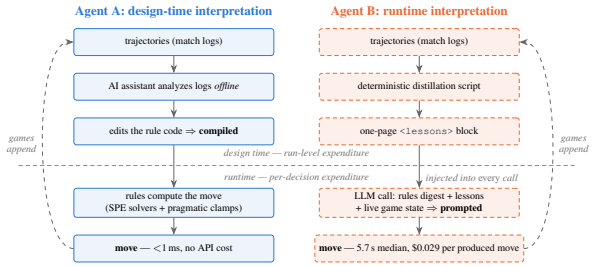


Figure 1: Two interpretation loops, consumed at opposite ends. Agent A's conclusions are compiled into the rules offline; nothing above the line runs when a move is due. Agent B's conclusions ride along as a prompt block, and the interpretation-bearing model call sits inside every decision.

Agent A: lessons compiled in. Bargaining uses subgame-perfect equilibrium (SPE) solvers, with backward induction on finite horizons and the Rubinstein closed form [8] on infinite ones. It infers the opponent's discount factor over the grid $\{0.7, 0.8, 0.9, 0.95, 1.0\}$, updated from observed offers and rejections. Negotiation uses reservation-value logic (never trade at a loss) with last-mover-aware surplus targeting and a concession curve. Persuasion builds on commitment-optimal disclosure [6]: lie on low-quality rounds at a design rate 25% below the buyer-indifference boundary (the commitment-optimal rate, above which trusting positive signals stops paying for the buyer). Pragmatic clamps (asks and thresholds capped at 85% of the pot) guard against spite-rejections. Every move is sub-millisecond. An AI assistant monitored ratings hourly and made $\sim$15 rule edits over ten days (Aug 11–21).

Agent B: lessons prompted in. Every decision is one call to an LLM (Anthropic predominantly `claude-opus-5`, alongside `claude-fable-5`, August-2026 snapshot, invoked headlessly at fixed CLI defaults with no exposed sampling parameters and a 90 s timeout) carrying a fixed per-family rules digest, the `<lessons>` block, and the live state compacted to the last six history entries plus whole-game aggregates. Lessons are deterministic aggregates of B's own trajectories (median opponent demands, acceptance rates, revealed seller honesty), updated through a fixed template without LLM judgment. Outputs are parsed as strict JSON and semantically validated; a timed-out, failed, or invalid call falls back to A's rule stack (0.9% of moves).

Competition refinement. A's development logs showed opponent demands of 53–57% and apparently unaccepted own offers. To make asks more sellable, a rule capped asks from round 3 at $\max(\text{SPE}, 0.58)$ before existing floors and clamps. A negotiation walk-away stop-loss (Aug 12–13, before the analysis window) was reverted after development notes reported lower payoff, restoring the option of later profitable deals. B's template turned rejected own offers in its logs into "price closer to the opponents' demonstrated band," hot-reloaded as soft guidance. B's median agreement round was 2 in the analysis window. These consequences document rule and prompt refinement, not isolated payoff effects. Both rejection-based examples used incomplete pre-action histories, audited in §3.3.

Games and metrics. Bargaining alternates offers over a pot; normalized payoff is own share. Negotiation exchanges prices between privately valued buyers and sellers; payoff is own profit divided by own valuation. Persuasion repeats quality signals and buy/pass decisions; payoff is normalized by price $\times$ rounds. Behavioral rates use eligible offers, agreements, or seller rounds as denominators. Values compare between agents within a family, not across families; metrics pool roles, configurations, and versions unless stated.

3 Results

Evolving deployments. These results pool configurations and policy versions, not fixed policies evaluated on a held-out set. A underwent repeated rule revisions. B logged three lessons versions (Aug 14, 14:08, 15:27, and 20:17), hence two changes, then fixed; 619 of 40,966 logged moves

carried the `claude-fable-5` label and the remainder `claude-opus-5`. Changes may overlap ongoing games; the pooled comparisons do not identify revision effects.

Table 1: Returns and costs. *Games* = games appearing in our trajectory logs within the frozen analysis windows; *payoff* n = the subset with authoritative results, the only games entering the payoff means and standard errors (timeout shares in Table 2). Normalized payoff ($\pm$ standard error) is per-game payoff normalized by pot (bargaining), own valuation (negotiation), or price$\times$rounds (persuasion); B$-$A differences in §3.1 are computed from unrounded means. Costs are totals over each agent's competition run and cover model-API expenditure only. [†]A's design-time expenditure was measured from local assistant-session transcripts at public API rates (the work ran on subscription usage). A's figure covers the build plus $\sim$240 hourly monitoring checks and rule edits over ten days; B's covers its build session (lessons updates used a deterministic script). B's runtime figure includes retries and failures. [‡]Upper bound: 1.34M A moves logged in the analysis window, at $<$1 ms each.

	Games		Payoff n		Norm. payoff	
Family	A	B	A	B	A	B
Bargaining	40,313	1,377	40,313	1,377	0.454 ± 0.001	0.461 ± 0.004
Negotiation	40,333	1,368	40,333	1,368	0.095 ± 0.001	0.095 ± 0.004
Persuasion	40,322	1,362	40,322	1,362	0.469 ± 0.003	0.465 ± 0.015
Cost: design time, run-level (AI build + analysis)[†]					\$151	\$86
decisions, runtime (marginal, model API)					none	\$1,158 (\$0.28/game)
inference time (total)					$<$0.4 h[‡]	80 h
latency per move (median)					$<$1 ms	5.7 s

3.1 Returns

Statistical note. Table 1 reports sample SD/$\sqrt{n}$ over games, treating games as independent. Repeated opponents and configurations are not clustered; deployments were unmatched and evolving. The standard errors therefore convey descriptive dispersion only; we make no hypothesis-test or equivalence claim. The B$-$A mean differences were +0.008, 0.000, and -0.005 in bargaining, negotiation, and persuasion, respectively.

Result coverage. An *authoritative result* is a terminal server outcome captured live or retrieved through the game-state API. The locally available retrieved outcomes now cover all logged games in both frozen windows, including 40 additional A outcomes since the original analysis. Normalized means at the reported precision are unchanged.

3.2 Costs: run-level versus per-decision

We price input, output, and cache tokens separately at August-2026 API rates, including retries and failures; human time, hosting, local compute, and deterministic scripts are excluded. B's decision layer cost \$1,158 over 63,238 calls; 40,609 produced moves (\$0.029/move, \$0.28/game); the rest were retries or failed attempts. Successful calls took 80 inference hours (median 5.7 s, p90 13.1 s). This bill covers the whole decision layer, not the lessons increment. A's run-level AI-assisted development and monitoring cost \$151 (\$0.001/game), with no runtime model expenditure; B's design cost was \$86. Scaling B to 120,968 games gives \$34k at fixed prices, prompt length, move counts, and retry rates (\$17k–\$68k if per-game cost halves or doubles).

3.3 Agent Behavior Analysis: Similar Point Estimates, Different Behaviors

Table 2 maps intended to observed behavior; the paragraphs below unpack the three largest divergences. Behavioral statistics use Table 1's frozen windows (A: Aug 14–25, once trajectory logging began; B: Aug 14–24).

Terminal outcomes change the acceptance picture. Our submitted analysis counted 66,411 own proposals (A: 65,327; B: 1,084): entries with identifiable proposer and gain in each game's last pre-action history, once per game and round. Acceptance was read from the same entry's `decision` field. All were rejections; accepted terminal offers were missing because no further action was due. Zero historical acceptances therefore did not mean zero accepted offers. Joining authoritative outcomes

Table 2: Intended vs. observed behavior.

	Agent A (compiled)	**Agent B (prompted)**
Intended	Clamped equilibrium strategy; no loss-making trades; lying 25% below the commitment-optimal rate.	Follow rules digest and lessons; maximize payoff.
Observed	As designed: high asks, margin-holding, 43% lying.	Settles earlier, thinner margin, lies 17%, withholds 36% of true recommendations.
Alignment	Code-level determinism (identical states, identical moves); high-level targets not separately verified.	Hard constraints (valid JSON, offers sum to pot, positive prices, legal decisions) enforced; soft guidance (loss-avoidance, lying ceiling) not guaranteed; lying audited post hoc.
Deviations	None at runtime; the *design* revised $\sim$15 times.	No briefed lying target; lied far below the briefed ceiling and A's design rate.
Assurance	Hourly rating checks; log review; deterministic replay.	Output validation, full trajectory logging; no guarantee of high-level policy adherence.
Failures	Rules cannot time out; game timeouts 0.23% (persuasion).	Failed/invalid/timed-out calls fall back to A's rules (0.9% of moves); game timeouts 2.35% (persuasion).

by game ID, `agreed_round` identifies the proposer (player 1 on odd rounds, player 2 on even rounds): A's own offers closed 17,732/39,961 (44.4%) agreements, B's 555/1,373 (40.4%). These fractions are shares of agreements. Including outstanding and final produced offers, deduplicated by game and round, corrects mean first observed asks to 74% (36,178 games) and 59% (1,027). Median agreement rounds were 3 and 2. Opponent asks sat modestly above even splits (medians 56% and 57%), compatible with fairness anchoring [4, 2]; strategic and prompt-induced explanations remain possible.

Two points on the volume–margin frontier. In negotiation, as seller, A agreed in 46% of games at a median agreed price of 1.19$\times$ its valuation; B agreed in 57% at 1.15$\times$. Across both roles, agreement rates were 49% (A, 40,333 games with authoritative results) and 56% (B, 1,368). The payoff point estimates were close in these unmatched logs (§3.1), despite visibly different operating points. Only the trajectories reveal that the systems reached different solutions.

The LLM lied far less than its brief allowed. Agent A lies on 43% of low-quality persuasion rounds (of 188,231 such rounds), its design rate 25% below the commitment-optimal boundary [6]. Agent B, briefed with a ceiling but no target rate, lied on only 17% (of 5,992), roughly 40% of A's rate, and withheld recommendations on 36% of its 7,446 high-quality rounds. The static model says B left profitable lies on the table; the cross-agent payoff difference does not establish an effect (§3.1); unmatched pools and no policy ablation preclude causal attribution.

4 Discussion

Competition lessons. First, similar payoff point estimates can conceal different agreement, margin, and persuasion strategies. Trajectory analysis helps reveal them, but must include terminal outcomes to avoid mistaking logging blind spots for opponent behavior. Second, compiled rules offer low runtime model expenditure and code-level predictability, but adapting them requires explicit revision. Third, runtime LLM control permits more discretion, incurs substantial inference expenditure, and gives no guarantee of following soft guidance. These unmatched, evolving runs do not establish whether that discretion improves payoff; matched-strategy ablations or offline replay are needed.

Limitations. The study covers one competition and one LLM family. Matchmaking, time, base strategies, iteration counts, and game volume differ between agents. Costs are model-API-equivalent estimates, excluding human time, hosting, and local computation; they are not a total-cost comparison. B's runtime bill covers the entire LLM decision layer, including retries and failures, rather than the incremental cost of adding the lessons block.

Data governance and availability. All statistics derive from our own agents' match records, collected via the competition API under its terms,[1] which expose no direct participant identifiers; the pool contains humans, so aggregates include games against them (anonymized by the platform). We do not analyze human-written messages in our raw logs and release only de-identified aggregates. Code, prompts, configurations, analysis scripts, and aggregated data are available at `https://github.com/starchou6/glee-iab-2026` (commit `cffe01f`).

[1] `https://glee-competition.com/terms`; the platform states the protocol was approved by the Technion ethics committee.

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